Employees Database Analysis (HR)

Employees Database Analytics: Project Report

1. Project Overview

This project involved a comprehensive analysis of an HR employees dataset. The primary objective was to clean uncleaned data and transform it for exploratory data analysis (EDA), identify employee details by region, create meaningful visualizations regarding employee status, experience, and performance, and provide actionable insights for management to improve employee satisfaction.

Objective(Problem statement)

1. To clean the uncleaned data and transform the data for analysis using exploratory data analysis (EDA)
2. To know about the employees details of a particular region
3. To create meaningful visualizations of the employees database to show the status, experience and Performance of every employees
4. To provide actionable insights and recommendations from the analysis

2. Data Sources

- Source: HR Employees Analysis dataset (GitHub Raw Data).
- Dataset Size: 1020 rows × 12 columns.
- Timeline: Data collected during 2020 to 2024.
- Domain: Human Resource.

3. Problem Statement

To analyse HR employee data by cleaning, transforming, and visualizing datasets to identify performance trends, salary distributions, and department-specific insights.

4. Attribute (Column/Features) Details

Attribute Name	**Data Type**	**Description**
Employee ID	String	Unique identifier for every employee.
Full Name	String	Combined first and last name.
Age	Float	Employee age (Range 25-40).
Department	String	Department identifier.
Region	String	Region identifier.
Status	String	Employment status (Active/Inactive/Pending).
Join Date	Date	Date of employee joining.
Salary	Float	Employee salary.
Email	String	Employee email address.
Phone	Integer	Contact number.
Performance Score	String	Employee performance assessment.
Remote Work	Boolean	Indicates if working remotely.

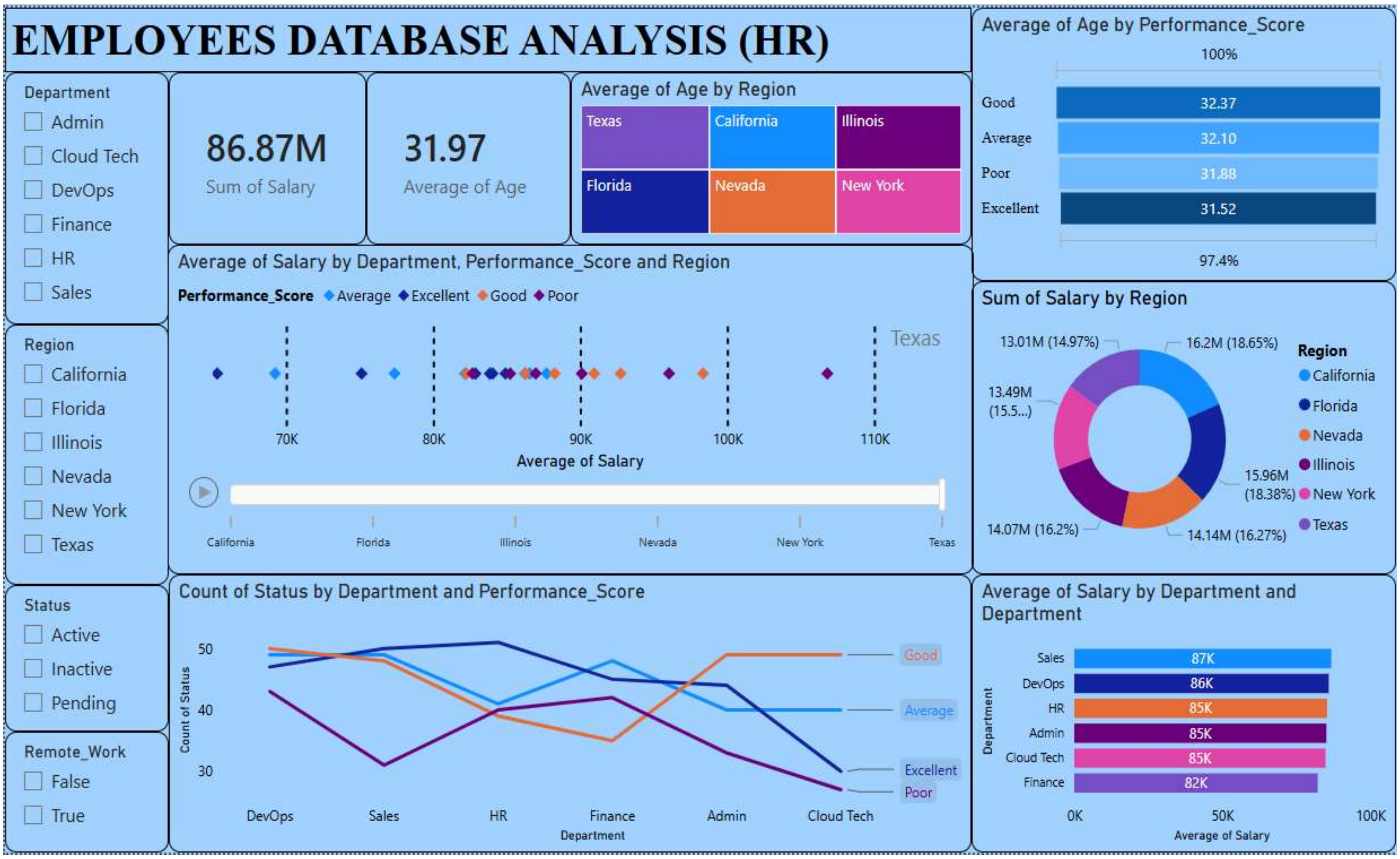
5. Tools & Technologies

- Programming Language: Python.
- Libraries: Pandas, NumPy, Matplotlib, Seaborn.
- Dashboard Creation : Microsoft Power BI Desktop

6. Data Pre-Processing

- Missing Values: Handled missing values in 'Age' and 'Salary' by imputing them with the median value.
- Duplicates: Verified the dataset for duplicate rows and confirmed none were present.
- Feature Engineering: Combined 'First Name' and 'Last Name' into a 'Full Name' column; split 'Department_Region' into separate 'Department' and 'Region' columns.
- Type Conversion: Converted 'Join_Date' to datetime objects and standardized format.
- Outliers: Statistical analysis using IQR confirmed no extreme outliers in 'Age' or 'Salary'.

7. Analysis and Visualizations



- **Age Distribution:** Employees are aged between 25 and 40, with the highest concentration at age 32.
- **Department Count:** The DevOps department has the highest workforce concentration, while Cloud Tech has the least.
- **Salary vs. Age:** There is no strong linear pattern suggesting salary strictly increases with age; rather, compensation is distributed across the age span.
- **Average Salary by Department:** DevOps and Sales show competitive salary levels, whereas Finance has the lowest average salary.
- **Average Age by Performance Score:** Most of the Employees Age lies between 31-32 and also excellent performer are at 32 and poor performer at 31 so there is no such age variation by performance .

8. Final Overall Insights

1. Total Expenditure: The organization spends 86.87 million on salaries.
2. Workforce Profile: The average age of employees is approximately 32 years, indicating a young and energetic team.
3. Regional Budget: California (18.65%) and New York (18.38%) combined account for over one-third of the total salary budget.
4. Performance & Pay: Excellent performance scores generally correlate with higher average salaries in certain departments, indicating the company rewards hard work.
5. Department Status: Departments like Admin and HR have even workforce distributions, whereas DevOps and Sales are primary drivers of salary expenditure.

9. Conclusion

This project successfully analyzed HR data to identify relationships between demographics, department performance, and salary compensation. The insights regarding regional budget allocation and the correlation between performance scores and salary provide a strong foundation for future management decisions, such as budget planning, hiring strategies, and retention programs for top-performing employees.